

YEREVAN ARABKIR, Armenia

WMO: 377890

Lat: 40.212N

Lon: 44.530E

Elev: 3325

StdP: 13.01

Time Zone: 4.00 (E04)

Period: 86-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	12.7	3.4	7.4	11.8	7.1	9.0	16.2	20.2	36.8	14.9	38.5	1.4	50	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	97.0	70.9	94.7	69.8	91.8	68.8	73.0	93.4	71.2	91.3	69.8	89.5	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	109.0	87.7	64.2	101.9	84.6	62.5	95.9	83.0	39.1	93.3	37.3	92.1	35.9	89.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.4	18.8	15.5	5.1	102.5	9.3	2.4	-1.6	104.2	-7.0	105.6	-12.2	107.0	-18.9	108.8		
			4.6	76.9	8.3	2.6	-1.4	78.8	-6.3	80.3	-11.0	81.8	-17.0	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	28.6	33.7	43.9	55.8	63.6	72.7	80.2	79.4	69.8	56.9	43.9	33.5
	DBStd	18.69	7.36	7.28	6.86	6.01	6.14	5.70	4.48	4.44	5.32	6.03	6.78	8.89
	HDD50	2068	662	456	207	16	0	0	0	0	0	13	203	511
	HDD65	4932	1127	876	654	282	103	8	0	0	16	257	634	975
	CDD50	3999	0	0	18	190	420	682	936	912	595	226	19	1
	CDD65	1388	0	0	0	5	58	240	471	447	162	5	0	0
	CDH74	16143	0	0	0	83	618	2628	5582	5257	1849	126	0	0
	CDH80	7470	0	0	0	13	139	1097	2857	2650	698	16	0	0
Wind	WSAvg	5.1	2.7	3.6	5.0	5.8	5.8	6.7	8.3	7.7	5.3	4.2	3.2	2.6
Precipitation	PrecAvg	22.7	1.7	1.8	2.3	3.0	3.1	2.0	1.5	0.9	0.9	1.8	1.7	1.4
	PrecMax	27.9	3.3	2.8	4.1	5.6	4.6	3.2	3.3	2.0	3.7	4.7	3.6	2.4
	PrecMin	17.8	0.6	1.0	0.7	1.5	1.9	0.4	0.1	0.3	0.1	0.6	0.4	0.6
	PrecStd	3.0	0.7	0.6	0.9	1.2	0.7	0.7	0.8	0.4	0.8	1.0	0.9	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.1	57.4	68.3	81.0	87.4	96.2	100.8	101.6	93.5	82.2	67.9	55.2
		MCWB	43.0	44.5	51.4	60.1	63.6	69.1	72.2	72.2	66.9	62.2	53.1	44.8
	2%	DB	47.6	53.6	64.3	76.1	84.0	92.8	97.3	97.1	89.9	77.3	63.0	50.6
		MCWB	39.7	42.4	49.3	57.2	62.2	68.0	71.1	70.9	66.5	59.6	50.9	43.5
	5%	DB	43.2	50.1	60.6	72.0	80.6	89.5	94.9	94.7	87.6	73.5	59.2	47.0
		MCWB	37.1	41.2	47.3	55.4	60.8	66.5	70.4	69.9	65.6	57.9	48.9	41.4
	10%	DB	39.5	46.3	57.2	68.4	77.2	86.3	92.4	91.7	84.2	70.0	55.7	44.2
		MCWB	34.8	39.2	46.3	53.5	59.2	65.4	69.3	69.1	63.8	56.1	47.3	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	46.2	53.0	61.6	65.7	71.8	75.6	75.3	70.1	63.6	54.3	47.9
		MCDB	50.8	53.8	64.6	77.9	83.7	92.7	95.3	95.9	88.1	76.8	64.1	53.1
	2%	WB	40.2	43.9	50.7	58.6	63.6	69.3	73.5	72.8	67.9	60.9	51.9	44.6
		MCDB	46.1	52.1	61.7	72.4	80.8	89.0	94.1	93.2	87.4	75.7	61.0	49.1
	5%	WB	37.6	41.6	48.4	56.5	61.9	67.6	71.6	71.2	66.1	58.7	49.9	41.7
		MCDB	43.0	49.2	58.3	69.3	77.6	87.4	91.7	91.6	85.2	71.6	58.1	46.2
	10%	WB	35.0	39.3	46.5	54.8	60.2	66.0	70.1	69.8	64.5	56.8	48.0	39.7
		MCDB	39.0	46.0	55.9	66.6	75.0	84.5	89.8	89.8	82.4	68.1	54.7	43.3
Mean Daily Temperature Range	5% DB	MDBR	15.4	18.5	21.2	21.8	22.9	25.3	24.5	25.2	26.8	23.5	21.1	14.7
		MCDBR	19.9	24.2	26.4	26.9	26.9	27.6	26.5	27.6	30.3	28.4	24.5	17.9
		MCWBR	13.4	15.5	14.9	13.1	12.0	11.3	10.9	11.3	13.3	14.4	14.4	11.6
	5% WB	MCDBR	18.6	22.0	23.7	23.2	24.7	26.7	25.4	26.2	29.3	26.6	22.9	15.8
MCWBR		12.7	14.3	14.5	12.6	11.9	12.0	11.6	12.0	13.7	14.3	13.7	10.7	
Clear-Sky Solar Irradiance	taub	0.301	0.335	0.391	0.457	0.430	0.411	0.455	0.439	0.378	0.375	0.320	0.297	
	taud	2.338	2.239	2.130	1.992	2.146	2.256	2.141	2.182	2.337	2.279	2.408	2.396	
	Ebn at Noon	272	276	271	259	270	275	261	262	273	259	263	265	
	Edh at Noon	28	36	44	55	48	43	48	45	36	34	26	24	
All-Sky Solar Radiation	RadAvg	725	1045	1227	1457	1799	2174	2175	1999	1631	1071	724	549	
	RadStd	82	143	119	170	102	133	103	108	100	80	97	76	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	+1.48	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page